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PAPER_1005: Yang-Mills Mass Gap via SCm BCS Phonon Coupling

Abstract

We compute the Yang-Mills mass gap Δ_{YM} through a BCS-like mechanism where the SCm phonon field provides the pairing interaction. The running coupling $\alpha_s(T)$ governs the gap magnitude above deconfinement.

1. Mass Gap Equation

$$\Delta_{\text{YM}} = \Lambda_{\text{QCD}} \cdot \exp(-1/(\alpha_s(T) \cdot N_c)) \cdot S_{26}^{(3)}$$

$$\alpha_s(T) = \alpha_{s,0} / (1 + \alpha_{s,0} \cdot b_0 \cdot \ln(T/T_c))$$

where $b_0 = (11N_c - 2N_f)/(12\pi)$.

2. Results

At $T = 2 \times 10^{12}$ K: $\Delta_{\text{YM}} = 9.96 \times 10^{-292}$ J. The extremely small value reflects deep asymptotic freedom at high temperature.

3. Implementation

File: scm_qgp_dynamics.py, class YangMillsMassGapCalc. CP4 class #589.